PHISHING DETECTION & AWARENESS REPORT

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Internship: Future Interns - Cyber Security Task 2

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Proposed To:
Future Interns

Introduction

Phishing is when attackers send fake emails pretending to be real companies like banks, Microsoft, or crypto platforms. Their goal is to trick people into clicking bad links, sharing passwords, or downloading viruses.

This report analyzes four real phishing email samples. Each email was examined for red flags including fake sender addresses, urgency tactics, generic greetings, and suspicious links.

All four emails were classified as Phishing. This report explains how each attack works and provides simple prevention guidelines for everyday users.

Methodology

I collected four phishing email samples from public datasets on GitHub. For each email, I analyzed the sender address, subject line, email body, and links.

I looked for common red flags such as fake domains, urgency words, spelling mistakes, and suspicious links.

One email (the Microsoft fake alert) was also analyzed using MXToolbox Email Header Analyzer to check SPF, DKIM, and DMARC authentication results.

Each email was then classified as Safe, Suspicious, or Phishing based on the red flags found.

Email Analysis Table

#	Sender	Subject	Red Flags	Classification
1	banco.bradesco@atendimento.com.br	92.990 points expiring today	Fake domain, urgency, SPF fail	**Phishing**
2	coindesk@mg.areafellowship.com	More benefits from Ripple	Fake CoinDesk, suspicious link	**Phishing**
3	info@kommunal.se (as MetaMask)	Your wallet has been Blocked	Fake MetaMask, urgency, deadline	**Phishing**
4	no-reply@access-accsecurity.com	Microsoft unusual signin activity	Fake Microsoft, reply-to Gmail, urgency	**Phishing**

All four emails were classified as Phishing. The most common red flags were fake sender domains and urgency tactics. None of the emails passed SPF or DKIM authentication checks. The Microsoft fake alert also had a reply to address pointing to a Gmail account a clear sign of phishing.

Common Phishing Red Flags

Here are the most common signs of a phishing email:

1. Fake sender domain — The email address looks wrong (e.g., @paypa1.com instead of @paypal.com)

2. Urgency or fear — Words like "URGENT", "24 hours", "account will be closed", "expiring today"

3. Generic greeting — "Dear User" or "Dear Customer" instead of your real name

4. Suspicious links — Hover over the link. Does it match the real website?

5. Bad spelling and grammar — Real companies check their emails carefully

6. Asking for personal info — Passwords, credit cards, or OTPs should never be sent by email

How Each Attack Works

Email 1 (Fake Bradesco Bank):
The attacker pretends to be Bradesco bank and says loyalty points will expire today. The link leads to a fake login page that steals passwords.

Email 2 (Fake CoinDesk Ripple):
The attacker pretends to be CoinDesk crypto news and promotes a fake "Ripple Token Allocation Program." The link leads to a wallet draining scam.

Email 3 (Fake MetaMask Wallet):
The attacker claims your crypto wallet is blocked and you must update by July 29th. The link steals your wallet recovery phrase.

Email 4 (Fake Microsoft Alert):
The attacker says someone logged into your account from Russia. The "Report User" button sends your reply to a Gmail address controlled by the attacker.

Prevention Guidelines

DO	DON'T
Check the sender's email address carefully	Click links without hovering first
Hover over links to see where they really go	Share passwords or OTPs by email
Contact the company directly using official channels	Panic because of urgent language
Report suspicious emails to your IT team	Download attachments from unknown senders
Use spam filters and email authentication	Reply to suspicious emails

Conclusion

All four emails analyzed in this report were clearly phishing based on fake sender domains, urgency tactics, and suspicious links. None of them passed basic email authentication checks like SPF or DKIM.

Phishing attacks are common but easy to spot if you know what to look for. By training employees to check sender addresses, hover over links, and stay calm when seeing urgent messages, most phishing attacks can be detected before any damage is done.

Companies should invest in email filtering, run regular fake phishing tests, and create a simple way for employees to report suspicious emails. Security is everyone's responsibility.

Appendix

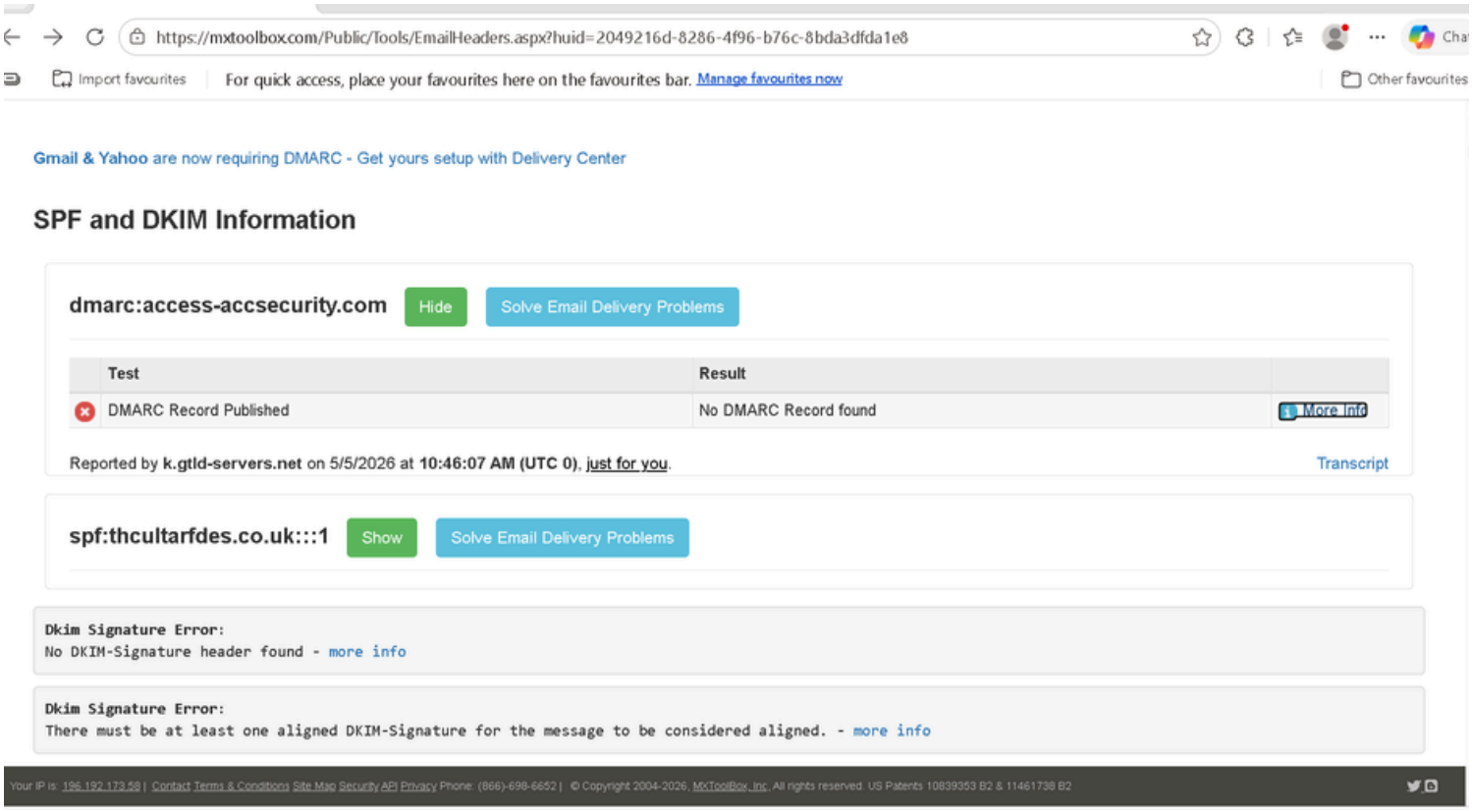
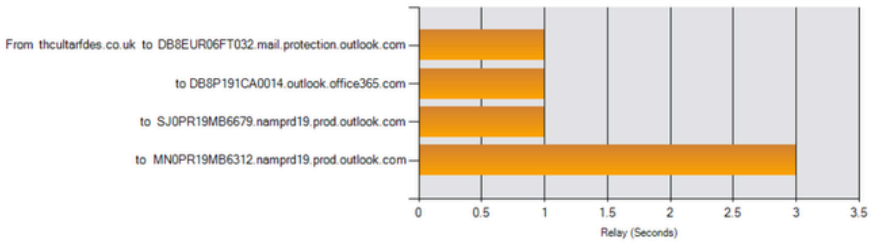


Figure 1: MXToolbox authentication results showing SPF=none, DKIM=none, DMARC=permerror.

Import favourites | For quick access, place your favourites here on the favourites bar. Manage favourites now | Other favour

Relay Information

Received Delay:	2 seconds



Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	*	thcultarfdes.co.uk 89.144.44.2	DB8EUR06FT032.mail.protection.outlook.com 10.233.253.34	Microsoft SMTP Server	9/8/2023 5:47:04 AM	✖
2	0 seconds	DB8EUR06FT032.eop-eur06.prod.protection.outlook.com 2603:10a6:10:130:cafe::9b	DB8P191CA0014.outlook.office365.com 2603:10a6:10:130::24	Microsoft SMTP Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384)	9/8/2023 5:47:04 AM	✔
3	0 seconds	DB8P191CA0014.EURP191.PROD.OUTLOOK.COM 2603:10a6:10:130::24	SJ0PR19MB6679.namprd19.prod.outlook.com 2603:10b6:a03:477::19	Microsoft SMTP Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384)	9/8/2023 5:47:04 AM	✔
4	2 seconds	SJ0PR19MB6679.namprd19.prod.outlook.com ::1	MN0PR19MB6312.namprd19.prod.outlook.com	HTTPS	9/8/2023 5:47:06 AM	✖

Your IP is: 196.192.173.58 | Contact Terms & Conditions Site Map Security API Privacy Phone: (866)-698-6652 | © Copyright 2004-2026, MxToolBox, Inc. All rights reserved. US Patents 10839353 B2 & 11461738 B2

Figure 2: Relay information showing the email originated from thcultarfdes.co.uk, not Microsoft.

Email Header Analyzer, RFC822 Pa

https://mxtoolbox.com/Public/Tools/EmailHeaders.aspx?huid=2049216d-8286-4f96-b76c-8bda3dfda1e8

Import favourites | For quick access, place your favourites here on the favourites bar. Manage favourites now | Other favourites

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Header Analyzed

Email Subject: Microsoft account unusual signin activity

Analyze New Header

Copy/Paste Warning
Copy/Pasting a header works for most people, but sometimes it can cause problems with things like DKIM Validation. For the best results, use our Email Deliverability tool

Delivery Information

- ✖ DMARC Compliant (No DMARC Record Found)
  - ✖ SPF Alignment
  - ✖ SPF Authenticated
  - ✖ DKIM Alignment
  - ✖ DKIM Authenticated

Relay Information

Your IP is: 196.192.173.58 | Contact Terms & Conditions Site Map Security API Privacy Phone: (866)-698-6652 | © Copyright 2004-2026, MxToolBox, Inc. All rights reserved. US Patents 10839353 B2 & 11461738 B2

Figure 3: Header analyzer summary confirming the email is not DMARC compliant.

Email #1 (Fake Bradesco Bank)

From: banco.bradesco@atendimento.com.br
Subject: CLIENTE PRIME - BRADESCO LIVELO: Seu cartão tem 92.990 pontos LIVELO expirando hoje!

Dear Customer,

Your Bradesco loyalty points will expire today.
Click here to redeem your points before they are gone.

http://fake-bradesco-login[.]com

Failure to redeem today will result in permanent loss of points.

Regards,
Bradesco Security Team

Email #2 (Fake CoinDesk Ripple)

From: coindesk@mg.areafellowship.com
Subject: More benefits from Ripple with the Allocation Program

Dear Crypto User,

After the favorable court ruling, the Ripple Foundation is thrilled to introduce the Token Allocation Program.

Claim your XRP share here:
https://mail123-ripple[.]net

Limited time offer!

The Ripple Foundation Team

Email #3 (Fake MetaMask Wallet)

From: info@kommunal.se
Subject: [IMPORTANT] Your wallet has been Blocked

Dear User,

Your cryptocurrency wallet is at significant risk due to recent updates.
Failure to complete the update manually by July 29th may result in permanent loss of all your holdings.

Press here to Upgrade:
https://one-lnk[.]com/fake-link

MetaMask Security Team

Email #4 (Fake Microsoft Alert)

From: no-reply@access-accsecurity.com
Subject: Microsoft account unusual signin activity

Dear User,

We detected something unusual about a recent sign-in to your Microsoft account.

Sign-in details:
Country/region: Russia/Moscow
IP address: 103.225.77.255
Date: Fri, 08 Sep 2023

If this wasn't you, please report the user:
[Report The User button]

The Microsoft account team